

Development

This project required undertaking several aspects of product development outside of the realm of pure computer science, such as electrical engineering and physical construction/design. I was largely in charge of those parts of the process, while also contributing foundational parts of the codebase that allowed our Arduino-based synth to convert input into output sounds.

For example, a specific feature that I developed was the “octave bend” mode, which allows the player to shift the octave of a key pressed by pushing the key left or right. This effectively triples the number of notes our synth-keyboard can play.

```
if (abs(yValue) > bendBuffer){  
  if (yValue < 0){  
    soundGen.setFrequency(keys[i].voice, noteFrequency(baseNote + i + 12));  
  }else{  
    soundGen.setFrequency(keys[i].voice, noteFrequency(baseNote + i - 12));  
  }  
}
```

This code snippet is the essential part of this feature. For each key, if the horizontal input is greater than a certain buffer the key's frequency will be adjusted by twelve notes (an octave) up or down based on the direction that the key is pushed.

Each physical key is actually a joystick with a 3D-Printed key attached to it. This allows us to easily send both vertical and horizontal input to the arduino-brain of our project that takes the input and generates the sound.



I modeled the final physical body of our instrument in Fusion 360 and took it to our Engineering facilities at RHS to laser cut the sides of our box. Using square stock, wood glue, and some screws, I assembled our final finished instrument case, in which we ultimately stuffed our wire jungle into, making our project semi-presentable.

Reflection and Takeaway

A major part of this project had to do with learning how to make software integrate with real hardware. We weren't just making a program, but software for a tool that people interact with physically. This not only made us familiarize ourselves with the electrical side of computing devices and product design, but also gave us a new appreciation of how the devices that we use each day interface with the user. Specific things I learned from this project were: laser cutting (have a look at the instrument's exterior), basic electronics and breadboarding (take off the lid of the Wamboard to take a peek), and a deeper understanding of music and sound generation; hit a note and hear it ring!

Wamboard

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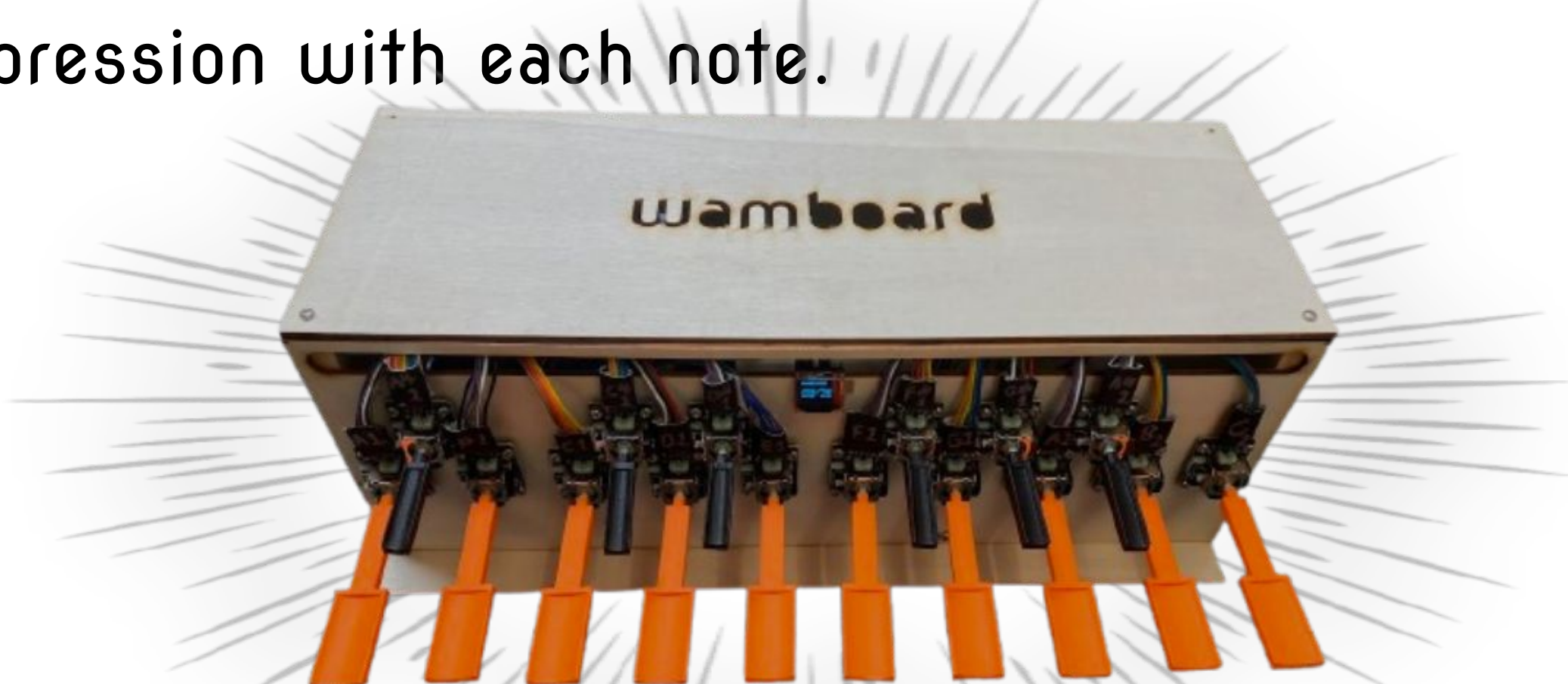
Roseville High, Computer Science,
Technology Innovation

What is it?

Do you play the piano?

Have you ever played the piano?

Well, if you have, you may have experienced the urge to **simply bend the key left and right to change the pitch of a note**. Imagine if each key was a whammy bar, allowing for satisfying expression with each note.



Dream no more! The *Wamboard* is exactly that. Each key has horizontal input, allowing you to alter the quality of each note played with a simple wiggle.

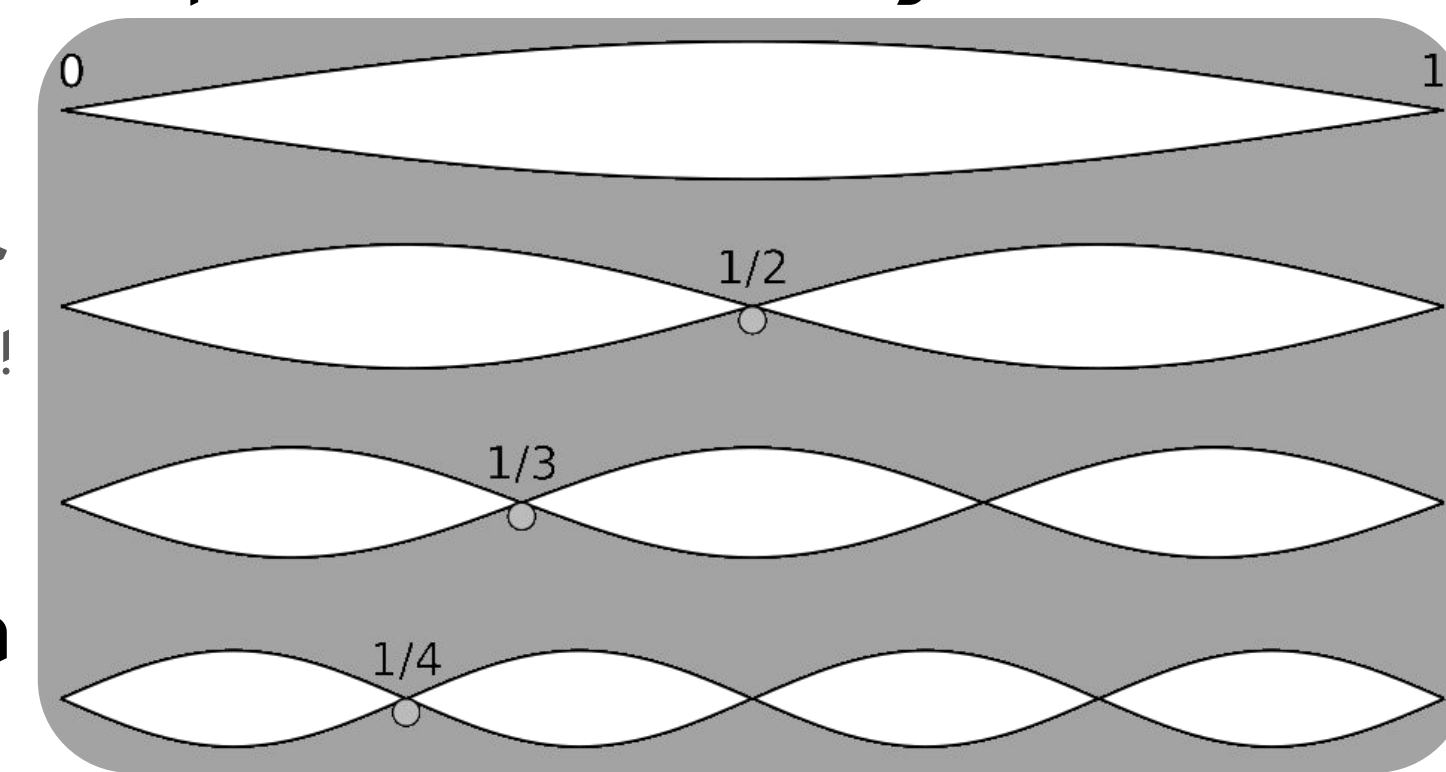


A Problem... And Solution!

We were pretty happy when we managed to make simple notes generated by our Arduino, but the notes sounded boring, and not particularly pleasant or interesting. This wasn't (entirely) reflexive of our ability to play the instrument, but largely due to the primitive and boring waveforms being played.

Overtones! →
Each is a higher pitch!

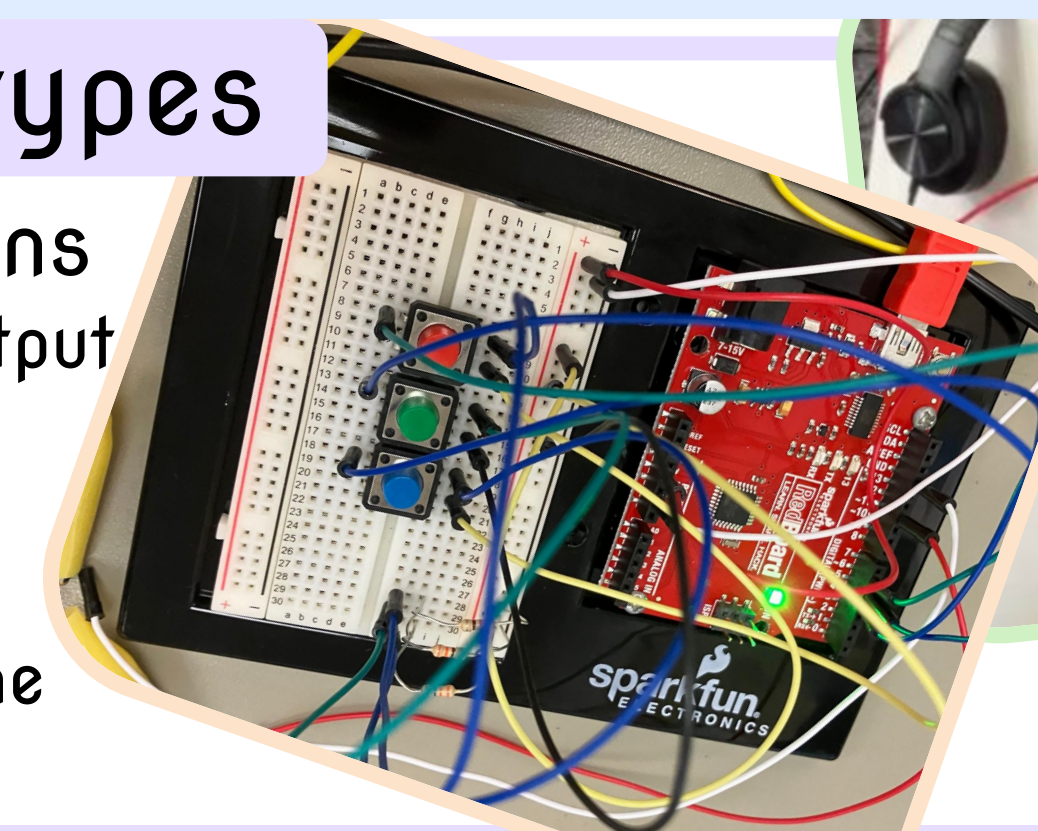
Any sound can be broken down into a set of combined sine waves of different frequencies. Musical instruments, unlike our instrument at the time, don't only produce one sine wave, but a set of sine waves of different frequencies. Every instrument has a different set of these that make their unique sound. These are called *overtones*.



To allow our instrument to play more complex and somewhat “realistic” instrument sounds, I created a Python script that takes in volume parameters for each overtone that should be added to the sound and outputs a wavetable (a set of numbers representing a sound wave that our synth can adjust and output as audio signals) generated from those overtones.

Prototypes

3 Buttons
Audio output involves touching wires to headphone jack.



Key Test
6 keys with horizontal and vertical input.

